A pruned spiking network as the first stage of a GRU-verified keyword spotter

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I. Aim

The requirement tests of the size and synapse study (R1) showed that no one-window network reaches 85 % live detection: running speech causes false accepts, and keeping them at 2 per hour costs about 20 points. The streaming cascade of this project (S1), a streaming SNN that proposes and a GRU network that verifies, meets the detection requirements on the same tests. This report asks whether our cheapest design, the pruned 48-neuron window SNN (W0: 0.35 ms, 12.4 KiB and 96.5 μJ per window), can replace the streaming first stage: does the cascade meet the requirements (Table II), at what cost, and does retraining the verifier on the window SNN's false proposals help? Table I lists the designs and their codes.

II. Architecture

Fig. 1 shows the cascade. Every 250 ms the window SNN scores the last 1 s window. When its margin reaches t_1 in this request or the previous one, the GRU verifier (Fig. 2) scores the last 1.5 s of 10 ms frames, and a head score of at least t_2 lights LED0. The decision logic is that of S1, unchanged; only the first stage differs. The two networks see the same log-mel bands: the window is 32 time bins of the 24 bands that the verifier reads frame by frame. Two first-stage scores were compared on validation: the single-window margin and the 2-of-3 confirmed margin of R1. Without a call limit they give the same validation recall, but the single-window margin needs fewer verifier calls; with the limit it detects more (85.6 against 78.1 %). It was used throughout.

III. Method

Harness. The cascade evaluation of the streaming design was copied unchanged; only its stage-1 scorer was replaced by the window SNN. Two checks came first. The harness's window features and integer model equal ours bit for bit (32,081 windows, 3 seeds). S1, rerun here, reproduces its published test result exactly (86.79 % live, 1.424 false accepts per hour, 0.07 % other words, 175 ms) and its validation selection. Our runs use the same audio and frames: window j ends in frame $98 + 25j$, inside request $j + 3$.

Thresholds. Both thresholds are chosen on validation with the rule used for S1: the highest live detection with at most 0.2 % other words and 1.6 false accepts per hour (a margin for 2 on test). Each design was then tested once. The second rule (A2, B2) adds at most 2,150 verifier calls per hour, the number S1 makes on validation. It was added after the test of A1, which showed about 10,000 calls per hour, and was fixed before its own test.

TABLE I
Designs compared. The letter gives the networks; the number gives the threshold rule (0: no verifier; 1: highest live detection; 2: at most 2,150 verifier calls per hour). The original verifier is the GRU network trained for the streaming design; the fine-tuned one was retrained here on the window SNN's false proposals.

Code	First stage (always on)	Verifier	Threshold rule
W0	window SNN (pruned, ours)	none	—
S0	streaming SNN	none	—
S1	streaming SNN	original	highest recall
A1	window SNN	original	highest recall
A2	window SNN	original	$\leq$2,150 calls/h
B1	window SNN	fine-tuned	highest recall
B2	window SNN	fine-tuned	$\leq$2,150 calls/h

TABLE II
Requirements and how they are measured. Every test set is the one used for the other designs in this report (the group's evaluation harness).

Requirement	Measured on	Target
Live "Sheila" detected	212 test clips in noise, 250 ms hop, 2 of 3 windows	$\geq 85\,\%$
Other words accepted	3,000 test words, same set-up	$\leq 0.2\,\%$
False accepts per hour	19.0 h: all other test words + LibriSpeech test-clean/-other	≤ 2
Word end to decision	median on a 1 h test stream, plus compute and link	≤ 500 ms
Compute per request	worst case on the board, 100 MHz	≤ 200 ms
Model memory	firmware model section	≤ 144 KiB
Average power, listening	PL, Vivado power $\times$ board cycles	≤ 1 W

Fine-tuning (B). The three window-SNN seeds scored 60 h of LibriSpeech training speech; their 17,725 strongest false proposals became the hard negatives of the original verifier's training recipe (Table III). Three verifiers were trained (about 3 h on the GPU) and combined with the three window-SNN seeds: nine cascades.

Cost. Worst compute per request is the stage-1 board time plus one verifier call (107.3 ms on plain `kdot`, from the RTL simulation of the parallel-processing design). Memory is the stage-1 model plus the 53,320-byte verifier. Power uses the Vivado estimates of the `ksdot` fabric. Busy-polling, the firmware as built, keeps the core running, so the power does not depend on the work. Clock gating, a calculation, adds the core energy of each request and each verifier call to a fixed floor.

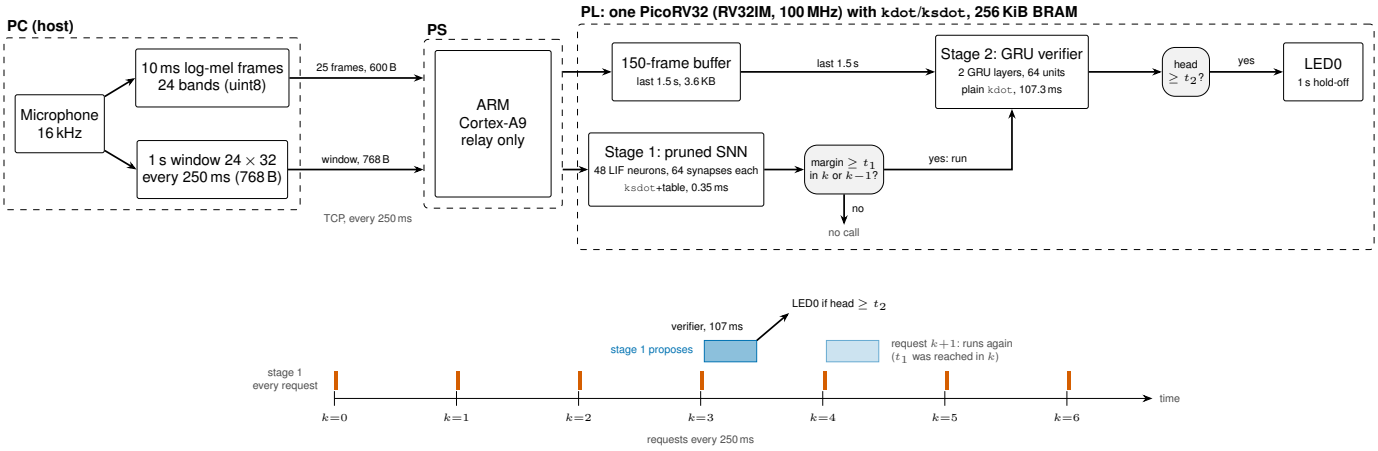


Fig. 1. Proposed cascade (evaluated in software with the board's integer models; no cascade has run on a board yet). Every 250 ms the PC sends the 1 s window for stage 1 and the 25 new 10 ms frames for the verifier's buffer. Stage 1, our pruned window SNN, costs 0.35 ms. When its margin reaches t_1 in this request or the previous one, the GRU verifier scores the last 1.5 s (107.3 ms on plain `kdot`); a head score $\geq t_2$ is a detection. The decision logic is that of the streaming design (S1), unchanged. Bottom: timing of one proposal.

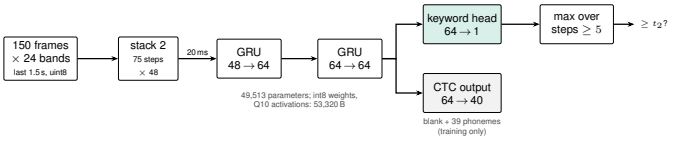


Fig. 2. The GRU verifier (stage 2), as trained for the streaming design. Two 10 ms frames are stacked into one 20 ms step; two GRU layers of 64 units read the last 1.5 s. A CTC phoneme output trains the layers on LibriSpeech; the keyword head, a linear readout per step, gives the decision: its maximum over the steps after a short warm-up is compared with t_2. The scores here come from its integer model, which its firmware reproduces bit for bit.

TABLE III
VERIFIER FINE-TUNE: THE ORIGINAL VERIFIER'S TRAINING RECIPE, UNCHANGED EXCEPT FOR THE STARTING POINT AND THE HARD NEGATIVES (WINDOWS WHERE THE WINDOW SNN PROPOSES "SHEILA" IN LIBRISPEECH TRAIN-CLEAN-100). SETTINGS READ FROM THE CHECKPOINTS.

Setting	Original verifier (A, S1)	Fine-tuned verifier (B)
Epochs × steps	10 × 500	10 × 500
Learning rate (one-cycle)	0.001	0.001
LibriSpeech per step	180 s	180 s
Keyword batch (keyword / near-miss share)	128 (0.25 / 0.15)	128 (0.25 / 0.15)
Hard windows per step	64	64
Loss weights (head / hard hinge / pos. hinge)	1 / 0 / 0	1 / 0 / 0
Real keyword clips only	yes	yes
Starting point	CTC base (`sheila_logmel`)	the original verifier
Hard negatives	8,000 from the streaming SNN	17,725 from the 3 window-SNN seeds
Seeds	1	3
Hardware	RTX 4060 Laptop (CUDA)	RX 7800 XT (ROCm, no MIOpen)

IV. RESULTS

Table IV gives every design against the requirements; Fig. 3 plots detection, Fig. 5 and Table V the cost.

Original verifier (A1, A2). A1 meets every requirement: 87.6 ± 0.3 % live, 1.83 false accepts per hour, against 36.3 % for W0. But the highest-recall rule sets t_1 below zero. The window SNN then wakes the verifier for most requests (10,244 calls per hour, against 2,222 for S1): the cascade reaches the verifier's own limit, which is also where S1 ends (87.75 % on validation for both). With at most 2,150 calls per hour (A2), live detection falls to 84.4 ± 0.9 %, 0.6 points short.

Fine-tuned verifier (B1, B2). Both rules now meet every requirement:

- *B2, call limit:* 85.8 ± 0.8 % live (7 of 9 cascades $\geq$85 %) with 1,908 calls per hour, 1.4 points above A2. Clip by clip, the nine cascades gain 32 keywords over A2 and lose 5; no single cascade differs significantly (exact McNemar, $p \geq 0.06$).
- *B1, highest recall:* 87.8 ± 1.0 % live (all 9 $\geq$85 %) with $3{,}044 \pm 984$ calls per hour, 70 % fewer than A1 at the same detection.
- Other words rise from 0.03–0.10 % to 0.15 %, still inside 0.2 % (Fig. 6).

The first stage. Fig. 4 shows why so many calls are needed: the window SNN ranks some real "Sheila" windows below much speech, so reaching them takes many proposals. The streaming SNN is the better proposer. Running speech remains the main source of false accepts in every design, W0 and S1 included (Fig. 7).

Cost. Every cascade meets all cost requirements: 107.6 ms worst compute (54 % of the budget), 64.5 KiB, 162–178 ms from word end to decision, and 0.367 W average PL power as built, half of the 0.727 W of S1, because the `ksdot` fabric is smaller than the dual-core fabric of S1. As built, the calls change nothing; with clock gating they would: 0.110 W at about 1,900 calls per hour against 0.177 W at 10,244. Counted with the method of the parallel-processing design, the work energy is 77 and 76 J per hour for A2 and B2, 120 J for B1 and 400 J for A1, against 277 J for S1.

TABLE IV

RESULTS ON THE TEST SETS (THRESHOLDS FIXED ON VALIDATION, TEST RUN ONCE). W0, A1, A2: MEAN OVER 3 WINDOW-SNN SEEDS; B1, B2: MEAN OVER 9 CASCADES (3 FINE-TUNED VERIFIERS × 3 WINDOW-SNN SEEDS). S1: THE STREAMING DESIGN'S FINAL TEST, REPRODUCED EXACTLY HERE; ITS MEMORY IS REPORTED WITH THAT DESIGN. POWER: BUSY-POLLING, AS BUILT (W0, A AND B ON THE KSDOT FABRIC, S1 ON THE DUAL-CORE FABRIC). √ MET, × MISSED.

Requirement	Target	W0 window SNN alone	A1 original verif. highest recall	A2 original verif. ≤2,150 calls/h	B1 fine-tuned highest recall	B2 fine-tuned ≤2,150 calls/h	S1 streaming SNN + orig. verif.
Live "Sheila" detected	≥85 %	36.3 ± 6.1 % ×	87.6 ± 0.3 % √	84.4 ± 0.9 % ×	87.8 ± 1.0 % √	85.8 ± 0.8 % √	86.8 % √
False accepts per hour	≤2	2.86 ×	1.83 √	1.09 √	1.48 √	1.21 √	1.42 √
Other words accepted	≤0.2 %	0.03 % √	0.10 % √	0.03 % √	0.15 % √	0.15 % √	0.07 % √
Word end to decision	≤500 ms	134 ms √	162 ms √	163 ms √	171 ms √	178 ms √	177 ms √
Compute per request	≤200 ms	0.35 ms √	107.6 ms √	107.6 ms √	107.6 ms √	107.6 ms √	117.3 ms √
Model memory	≤144 KiB	12.4 KiB √	64.5 KiB √	64.5 KiB √	64.5 KiB √	64.5 KiB √	—
Average PL power	≤1 W	367 mW √	367 mW √	367 mW √	367 mW √	367 mW √	727 mW √
Verifier calls per hour		0	10,244	1,937	3,044	1,908	2,222
Live, complete recordings		42.7 %	92.9 %	90.2 %	93.6 %	91.8 %	92.0 %
Stream recall (1 h)		47.4 %	87.9 %	85.7 %	89.5 %	88.0 %	87.9 %

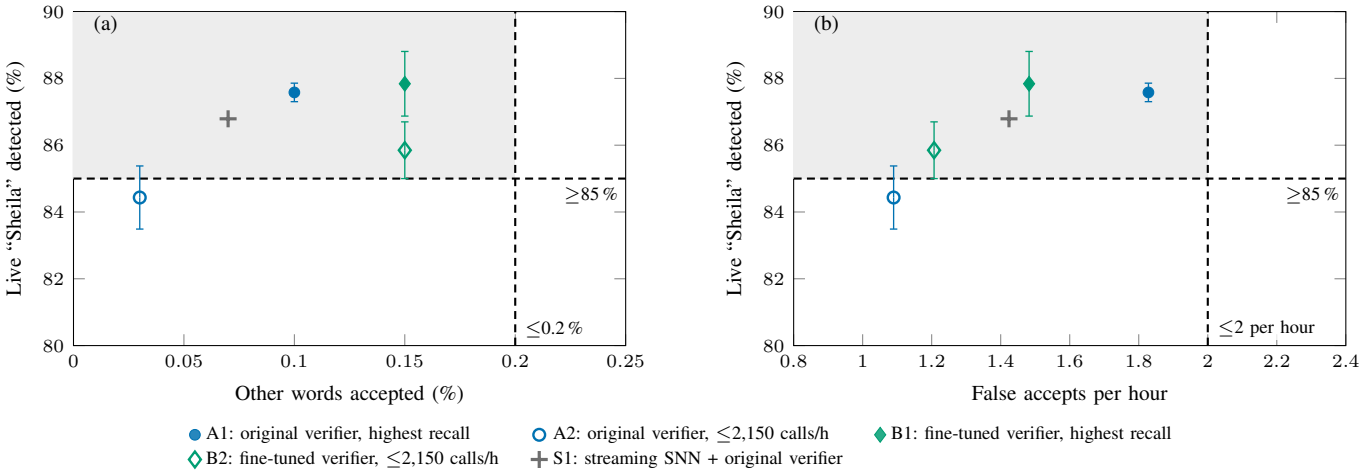


Fig. 3. Detection quality of the cascades on the test sets against the requirements; the shaded corner meets both. Each threshold was fixed on validation and the test was run once. Mean ± sd: 3 window-SNN seeds (A1, A2), 9 cascades for B1 and B2 (3 fine-tuned verifiers × 3 window-SNN seeds); S1 is a single model. Colour: the verifier (blue: original; green: fine-tuned on the window SNN's false proposals). Filled: the highest-recall rule; hollow: at most 2,150 verifier calls per hour. Off this scale, the first stages alone: W0 (window SNN) 36.3 % live (0.03 % other words, 2.86 per hour) and S0 (streaming SNN) 63.7 % (0.00 %, 1.58 per hour).

TABLE V

COST BREAKDOWN. E_V: PL ENERGY PER VERIFIER CALL (TOTAL PL POWER × 105.83 MS); WORK: 14,400 STAGE-1 REQUESTS PLUS THE VERIFIER CALLS PER HOUR, BOTH AT TOTAL PL POWER (THE METHOD OF THE PARALLEL-PROCESSING DESIGN). BUSY: AVERAGE PL POWER AS BUILT. GATED: WITH CLOCK GATING, A CALCULATION. W0, A AND B: KSDOT FABRIC (0.367 W); S1: DUAL-CORE FABRIC (0.727 W, 7.40 MJ PER STAGE-1 REQUEST).

Design	$t_{\max}$ (ms)	Memory (KiB)	Calls per h	E_V (mJ)	Work (J/h)	Busy (mW)	Gated (mW)
W0	0.35	12.4	0	—	2	367	94
A1	107.6	64.5	10,244	38.8	400	367	177
A2	107.6	64.5	1,937	38.8	77	367	110
B1	107.6	64.5	3,044	38.8	120	367	119
B2	107.6	64.5	1,908	38.8	76	367	110
S1	117.3	—	2,222	76.9	277	727	—

V. TRADE-OFFS AND KEY FINDINGS

- **The cheapest first stage works with a verifier.** The window SNN plus the GRU verifier meets all seven requirements, from 36 % live detection alone.
- **The verifier does the detecting.** Without a limit, the window SNN becomes almost a pass-through; the cascade's detection is the verifier's.
- **B1 and B2, neither preferred.** B2 uses 37 % fewer verifier calls (1,908 against 3,044 per hour) but sits 0.85 points above 85 %, with 2 of 9 cascades below; B1 keeps a 2.8-point margin. On today's board both draw the same power; with clock gating B2 would save 9 mW (0.110 against 0.119 W).
- **Fine-tuning helps where calls are limited.** It raises detection at a fixed number of calls, or cuts the calls at a fixed detection; it does not raise the verifier's own limit.
- **Energy needs idle hardware.** With busy-polling, every verifier call is free. Fewer calls save energy only when the core can stop between requests.

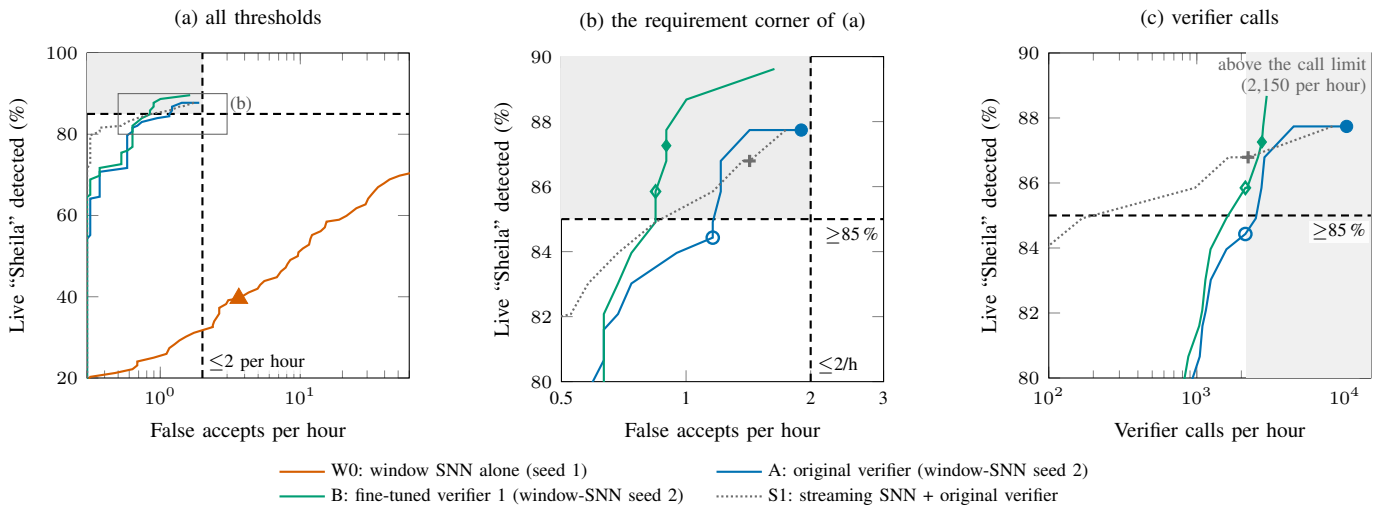


Fig. 4. What each verifier can trade, on test. Lines: every stage-1 threshold t_1 at the verifier threshold t_2 fixed on validation; markers: the operating points fixed on validation (styles as in Fig. 3). The configuration shown sits at the design means. (a) All thresholds; (b) the requirement corner of (a); (c) live detection against verifier calls, only for thresholds that also meet ≤0.2 % other words and ≤2 false accepts per hour. On test, the streaming SNN gives 84 % with at most 120 verifier calls per hour and 85 % from 971; the window SNN needs 2,129 calls per hour for 85 % with the fine-tuned verifier and 2,735 with the original one. Illustration only: no operating point was chosen on these curves.

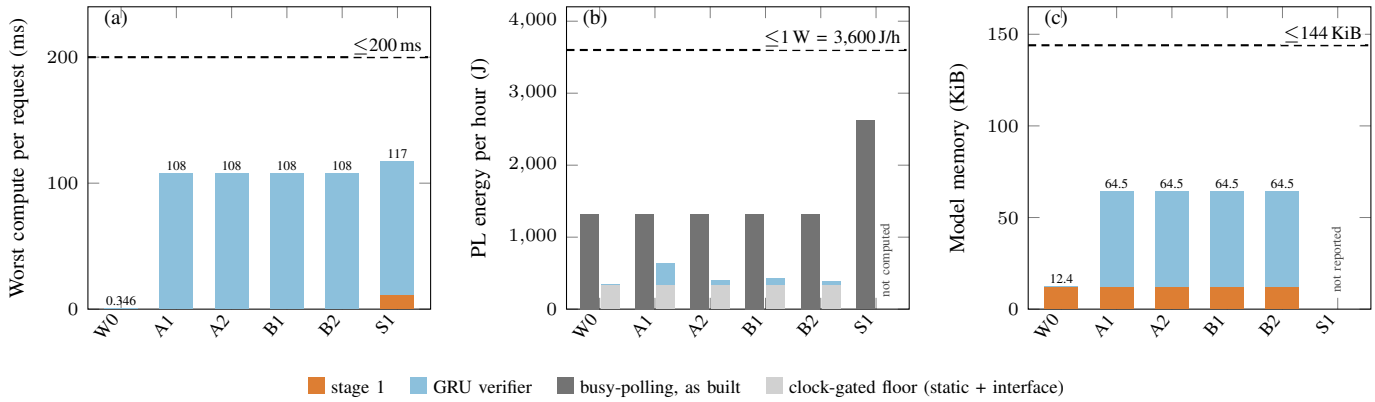


Fig. 5. Cost of the cascades. (a) Worst compute per 250 ms request: the window SNN on `ksdot`+table (0.35 ms, from board runs) plus one verifier call on plain `kdot` (107.3 ms, from the RTL simulation of the parallel-processing design); S1 as reported for the streaming design. (b) PL energy per hour: left bar, busy-polling (the firmware as built: the core never idles, so the energy does not depend on the work); right bar, clock gating (a calculation): the floor plus the core energy of stage 1 and of the verifier calls. W0, A and B on the `ksdot` fabric; S1 on the dual-core fabric of the parallel-processing design (0.727 W). (c) Model memory: stage-1 model plus the 52.1 KiB int8 verifier.

VI. Later work

- **Clock gating (next step).** First a measured number without RTL changes: lower the FPGA clock from Python and read the USB meter; then stop the core and BRAM clocks between requests.
- **Hardware merge.** Run the cascade on the board: the PC also sends the 10 ms frames, the core keeps a 1.5 s buffer, and the GRU instructions of the parallel-processing design cut a verifier call from 105.83 to 41.16 ms (they clash with `ksdot` on the opcode and must move).
- **A better proposer.** Retrain the window SNN on the streaming design's training data to cut running-speech proposals.

Limits. Three window-SNN seeds and three verifier seeds. Live detection uses 212 test keywords (one clip is 0.47 points), and 2 false accepts per hour over 19 h are about 38 events. Calls are counted on a speech-dense stream, so quiet rooms need fewer. Power is a Vivado estimate, and no cascade has run on a board. GPU training is not bit-reproducible; the seeds show the spread. The verifier's training recipe uses only the 4,218 of 32,000 synthetic training words that have dictionary pronunciations.

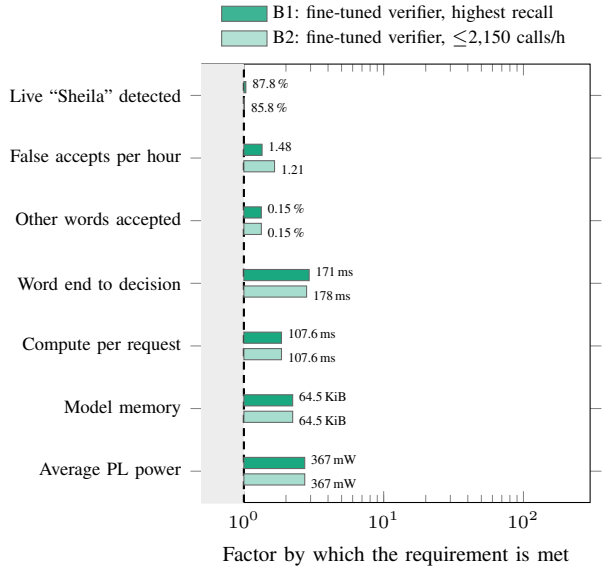


Fig. 6. B1 and B2 against every requirement, as the factor by which each is met (value/target for live detection, target/value otherwise; mean of 9 cascades); numbers are the measured values. Both meet all seven. Live detection and other words are the tight ones: 1.01 and 1.3 times for B2.

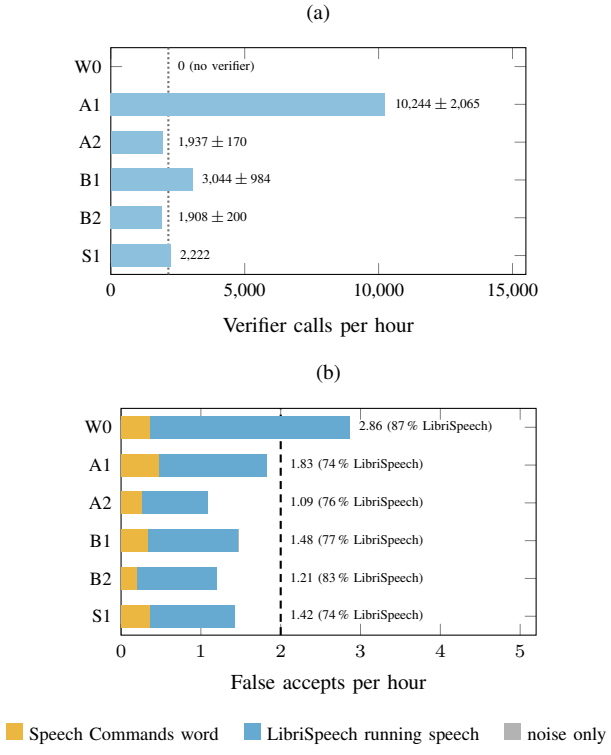


Fig. 7. (a) Verifier calls per hour on the 19 h negatives stream (mean ± sd; dotted: the 2,150 call limit). (b) False accepts per hour on the same stream by what was playing in the second before the detection (dashed: the requirement). Running speech is most of the false accepts in every design.